INCH-POUND

MIL-V-29601(AS)
15 December 1994

MILITARY SPECIFICATION

VEST, SURVIVAL, CMU-30/P22P-15

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements and inspections for one type of survival vest designed to accommodate soft and hard body armor.

1.2 Classification. The Aircrew Integrated Survival Armor Protection (AISAP) vests shall be one of the following sizes (see 6.2b):

Medium
Large

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2e).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Systems Standardization, Code 4.1.11B120-3, Naval Air Warfare Center Aircraft Division, Highway 547, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal(DD Form 1426) appearing at the end of this document or by letter.

AMSC-N/A FSC 8415

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SPECIFICATIONS

FEDERAL

V-F-106	-	Fastener, Slide, Interlocking
UU-P-268	-	Paper, Kraft, Wrapping
PPP-T-45	-	Tape, Gummed, Paper Reinforced and Plain, for Sealing and Securing

MILITARY

MIL-R-3390	-	Ring, Dee
MIL-C-3953	-	Cloth, Duck, Nylon (Validated)
MIL-W-4088	-	Webbing, Textile, Woven Nylon
MIL-T-5038	-	Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-W-5664	-	Webbing, Textile, Elastic (Validated)
MIL-C-7219	-	Cloth, Duck, Nylon
MIL-K-8662	-	Knife, Hunting, Sheathed, Survival, Pilot's (Validated)
MIL-B-17757	-	Boxes, Shipping, Fiberboard (Modular Sizes)
MIL-F-21840	-	Fastener Tapes, Hook and Pile, Synthetic

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MS20230	-	Grommet, Metallic, Plain and Spur, with Washer, Type I and Type III (Validated and Reinstated)
MS22018	-	Snap, Parachute Harness Quick Fit Ejector
MS22020	-	Link, Parachute Harness Triangle (Validated)
MS27980	-	Fastener, Snap, Style 2 (Regular Wire Spring Clamp Type)
MS27983	-	Fastener, Snap, Style 4 (Validated)
MS51940	-	Loop, Slide (For Equipage)

(Unless otherwise specified, copies of federal and military specifications and standards are available from DODSSP, Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094).

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2c).

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NAVAL AIR SYSTEMS COMMAND DRAWINGS

823AS100	-	Dee-Ring Assembly, Snaplink, Rescue
3241AS301	-	Survival Vest Assembly, CMU-30/P22P-15
3241AS302	-	Survival Vest Sub-Assembly, CMU-30/P22P-15
3241AS303	-	Securing Belt Assembly, CMU-30/P22P-15
3241AS304	-	Torso Harness Assemblies, CMU-30/P22P-15
3241AS305	-	Flashlight Strap Assembly, CMU-30/P22P-15
3241AS306	-	Chemlite Pocket Assembly, CMU-30/P22P-15
3241AS307	-	Heed Pocket Assembly, CMU-30/P22P-15
3241AS308	-	Survival Item Pocket Assembly, CMU-30/P22P-15
3241AS309	-	Day/Night Flare Pocket Assembly, CMU-30/P22P-15
3241AS310	-	Sea Dye Marker Pocket Assembly, CMU-30/P22P-15
3241AS311	-	Survival Knife Pocket Assembly, CMU-30/P22P-15
3241AS312	-	Pengun Flare Signal Kit Pocket CMU-30/P22P-15
3241AS313	-	SRU-31/P Medical Pocket Assembly, CMU-30/P22P-15
3241AS314	-	SDU-5/5 Light Storage Pocket Assembly, CMU-30/P22P-15
3241AS315	-	AN/PRC Radio Pocket Assembly, CMU-30/P22-15
3241AS316	-	D-Ring Retaining Flap Assembly, CMU-30/P22P-15
3241AS317	-	Identification Label, CMU-30/P22P-15
3241AS390	-	Mesh Vest Patterns, Size Medium, CMU-30/P22P-15
3241AS391	-	Mesh Vest Patterns, Size Medium, CMU-30/P22P-15
3241AS392	-	Pocket Patterns, CMU-30/P22P-15

(Unless otherwise specified, copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer.)

2.2 Non-Government publication. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified issues of documents

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not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2e).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D3951 - Packaging, Commercial

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The vests furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.3 and 6.4). Qualification of any one size assembly, under this specification, will automatically qualify a supplier for all sizes specified herein (see 1.2).

3.2 First article. When specified (see 6.2f), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.4.

3.3 Materials and components. The materials and components shall conform to applicable specifications, standards, and drawings and shall be as specified herein. Equivalent materials and parts require approval by the Naval Air Warfare Center Aircraft Division Warminster (Code 603), Warminster, Pennsylvania 18974-0591. Requests for use of equivalent items shall be submitted to the Contracting Officer.

3.3.1 Mesh Vest. The mesh vest shall be fabricated in accordance with Naval Air Warfare Center Aircraft Division Warminster Purchase Description and match Olive Green 106 (see 3.4.1.2).

3.3.2 Vest components. The vest components (pockets, flaps, liners, patches) shall be fabricated from aramid oxford cloth conforming to MIL-C-7219, Color 106 Olive Green (see 3.4.1.2).

3.3.3 Stiffener. The stiffener shall be fabricated from nylon duck cloth conforming to MIL-C-3953, Color 106 Olive Green (see 3.4.1.2).

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3.3.4 Webbings.

3.3.4.1 Survival vest sub-assembly webbing. The webbing used to fabricate the survival vest sub-assembly shall be fabricated from nylon conforming to MIL-T-5038, Type IV, Class II, 2 inch wide as specified in drawing 3241AS302.

3.3.4.2 Securing belt assembly webbings. The webbings used to fabricate the securing belt assembly shall be fabricated from nylon conforming to MIL-W-4088, Type VIII, Class I, 1 23/32 inch wide and Type IV, Class 1, 3 inch wide as specified in drawing 3241AS303.

3.3.4.3 Torso harness assembly webbings. The webbings used to fabricate the torso harness assembly shall be fabricated from nylon conforming to MIL-W-4088, Type XIII, Class I, 1 23/32 inch wide and from elastic conforming to MIL-W-5664, Type I, Class I, 1 1/2 inch wide as specified in drawing 3241AS304.

3.3.4.4 Flashlight strap assembly webbing. The webbing used to fabricate the flashlight strap assembly shall be fabricated from elastic conforming to MIL-W-5664, Type I, Class III, 1 1/2 inch wide as specified in drawing 3241AS305.

3.3.4.5 HEED pocket assembly webbing. The webbing used to fabricate the HEED pocket assembly shall be fabricated from elastic conforming to MIL-W-5664, Type I, Class III, 1 inch wide as specified in drawing 3241AS307.

3.3.4.6 AN/PRC radio pocket assembly webbing. The webbing used to fabricate the AN/PRC radio pocket assembly shall be fabricated from elastic conforming to MIL-W-5664, Type I, Class III, 2 inch wide as specified in drawing 3241AS315.

3.3.5 Tapes.

3.3.5.1 Tape. The tape used to fabricate the survival vest assembly shall be fabricated from nylon conforming to MIL-T-5038, Type III, Class II, 1 inch wide as specified by the applicable drawings in 2.1.2.

3.3.5.2 Seam tape. The seam tape used to fabricate the survival vest assembly shall be fabricated from nylon conforming to MIL-T-5038, Type III, Class II, 1 inch wide as specified by the applicable drawings in 2.1.2.

3.3.5.3 Hook tapes. The hook tapes used to fabricate the survival vest assembly shall be fabricated from nylon conforming to MIL-F-21840, Type II, Class I, 1, 1 1/2 and 2 inch wide as specified by the applicable drawings in 2.1.2.

3.3.5.4 Pile tapes. The pile tapes used to fabricate the survival vest assembly shall be fabricated from nylon conforming

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to MIL-F-21840, Type II, Class I, 1, 1 1/2, 2, and 4 inch wide as specified by the applicable drawings in 2.1.2.

3.3.6 Snap fasteners.

3.3.6.1 Eyelets. The eyelets used to fabricate the survival vest assembly shall be in accordance with MS27983-4 and MS27980-8B as specified by the applicable drawings in 2.1.2.

3.3.6.2 Studs. The studs used to fabricate the survival vest assembly shall be in accordance with MS27983-3 and MS27980-7B as specified by the applicable drawings in 2.1.2

3.3.6.3 Sockets. The sockets used to fabricate the survival vest assembly shall be in accordance with MS27983-2 and MS27980-6B as specified by the applicable drawings in 2.1.2.

3.3.6.4 Buttons. The buttons used to fabricate the survival vest assembly shall be in accordance with MS27983-1 and MS27980-1B as specified by the applicable drawings in 2.1.2.

3.3.7 Slide fasteners.

3.3.7.1 Survival vest sub-assembly slide fastener. The slide fastener used to fabricate the survival vest sub-assembly shall be in accordance with V-F-106 as specified in drawing 3241AS302.

3.3.8 Slide Loops. The slide loops used to fabricate the survival vest sub-assembly shall be in accordance with MS51940-9S as specified in drawing 3241AS302.

3.3.9 D-rings.

3.3.9.1 Survival vest sub-assembly D-rings. The D-rings used to fabricate the survival vest sub-assembly shall be in accordance with MIL-R-3390, Class 2, as specified in drawing 3241AS302.

3.3.9.2 Torso harness assembly gated D-ring. The gated D-ring used to fabricate the torso harness assembly shall be in accordance with drawing 823AS100, as specified in drawing 3241AS304.

3.3.10 Grommets. The grommets used to fabricate the survival vest assembly shall be in accordance with MS20230B20 as specified by the applicable drawings in 2.1.2.

3.3.11 Triangle parachute Link. The triangle parachute link used to fabricae the torso harness assembly shall be in accordance with MS22020-1 as specified in drawing 3241AS304.

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3.3.12 Parachute snap. The parachute snap used to fabricate the torso harness assembly shall be in accordance with MS22018 as specified in drawing 3241AS304.

3.3.13 Knife sheath. The knife sheath shall conform to MIL-K-8662, except that it shall not contain the sharpening stone, sharpening stone pocket, nylon laces, lace holes and slots for insertion of belt as specified in drawing 3241AS311.

3.4 Design and construction. Unless otherwise specified, the design and construction of the vests shall conform to the drawings listed in 2.1.2 of this specification.

3.4.1 Cut edges. The cut edges of the nylon webbings and tapes, except for the hook and pile fastener tapes, shall be seared, prior to the fabrication of the vests, to prevent fraying. No sharp edges shall be formed. The cut edges of the nylon duck cloth, except those that are completely covered by nylon binding tape, hook and pile fastener tapes, or are completely encased by the seam construction, shall be overedged stitched (serged).

3.4.1.1 Patterns. Standard patterns will be furnished by the Government to the contractor for use in cutting working patterns (see 6.6). The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way. Seams shall be in accordance with seam allowances specified in 3.4.

3.4.1.2 Component parts. The component parts shall be cut from the material specified, in accordance with the pattern parts in Table I.

3.4.2 Seams and stitching. All the seams used in the fabrication of the vests shall conform to the applicable drawings. Except for the serging (overedge) and zig-zag stitching, the stitching in the vest shall be accomplished with Stitch Type 301. The overedge stitching shall be accomplished with Stitch Type 502, 503, 504, or 505, and the zig-zag stitching shall be accomplished with Stitch Type 304. The width of the overedge and zig-zag stitching shall be approximately 1/4 inch. The thread and stitching used in fabrication shall conform to applicable drawing requirements. All the seam edges shall be properly forced out and shall not contain any folds. No seams shall be twisted, puckered, or pleated and no part of the vest shall be caught in an unrelated operation or seam. All the thread ends shall be trimmed to a length of 1/4 to 3/8 inch and except for the webbing, cord, and tape, excluding the binding tape, no raw or cut edge shall be visible on the exterior of the vest. The cut edges of the binding tape shall be turned under.

3.4.2.1 Sewing. Each row of stitching shall be straight and parallel to seam edge. The straightness of the stitching, in any row, shall be maintained within a tolerance of plus or minus

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1/32 inch. The ends of the stitching that are not caught in other seams or stitching shall be securely backstitched, by overlapping on themselves, for at least 1/2 inch. Thread breaks, skips, and run-offs shall be over-stitched not less than one inch. The thread tension shall be maintained so that there shall not be any loose or tight stitching and the lock shall be embedded in the materials sewn together.

3.4.2.2 Bartacking. Unless otherwise specified, bartackings shall conform to applicable drawing requirements.

3.5 Identification labels. Unless otherwise specified, identification labels shall conform to the requirements of Dwg 3214AS317.

3.5.1 Serial numbers. The vests shall be identified by individual serial numbers which shall be assigned by the manufacturer. Serialization shall be by a block of consecutive numbers to cover the entire acquisition document quantity.

3.6 Markings. Unless otherwise specified, the markings shall be legible and durable upper case black letters and Arabic numerals and shall be thoroughly dry prior to packaging. The markings shall be lusterless and the letters and numerals shall be sharply defined. The markings shall be accomplished by using such methods and materials as shall be resistant to erasure or fading under the condition of usage. The marking medium shall not contain any materials which are deleterious to the substance being marked.

3.7 Color. Unless otherwise specified, the color of the vests and components shall conform to applicable drawing requirements.

3.8 Visual examination. When visually examined as specified in 4.7.1, the vests shall conform to the requirements of this specification and applicable drawings. Table II, Classification of Defects for Visual Examination, shall be used to classify the defects.

3.9 Dimensional check. When dimensionally checked as specified in 4.7.2, the dimensions of the assembled vests, the dimensions and locations of components, the widths of seams and hems, and the stitching shall conform to applicable drawing requirements. Table III, List of Defects for Dimensions shall be used to enumerate the defects.

3.10 Packaging. When inspected as specified in 4.7.3, the fully prepared shipping containers, containing vests, shall conform to the requirements of this specification. Table IV, List of Defects for Packaging, shall be used to enumerate the defects found.

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3.11 Workmanship. After completion of the final assembly, the vests shall be thoroughly cleaned and all loose thread, lint, and foreign matter shall be removed. The rivets, grommets, and snap fasteners shall be clinched without distortion, damage, splitting, or cutting of the cloth, webbing, or tape, as applicable. The metal components shall not be misaligned nor contain any sharp edge, crack, dent, nick, burr, or sliver. The vests shall not contain any non-specified hole, abraded area, tear, cut, mend, needle chew, spot or stain. Drill holes shall be not greater than 1/16 inch in diameter and shall be covered by the component assemblies. The vests shall conform to the quality and grade of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. First article inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Quality conformance verification inspection (see 4.6)

4.3 Qualification inspection. Qualification inspection of the vests shall be as specified in Table V.

4.3.1 Qualification samples. Qualification samples shall consist of three vests (size optional to the manufacturer).

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Samples shall be forwarded to the test facility set forth in the letter of authorization to submit samples (see 6.4). The samples shall be plainly identified by securely attached durable tags marked with the following information:

> Samples submitted by (Name) on (Date) for qualification inspection in accordance with the requirements of MIL-V-29601(AS) and qualification test number ( ) under authorization (reference authorizing letter).

4.3.2 Retention of qualification. The retention of qualification shall consist of periodic verification to determine compliance of the qualified vests with the requirements of this specification. The time and method of periodic verification shall be specified by the activity responsible for the Qualified Products List and shall be included in the Notice of Qualification letter.

4.4 First article inspection. The first article inspection of the vests shall be as specified in Table VI.

4.4.1 First article samples. Unless otherwise specified, as soon as practicable after award of the contract or order, the contractor shall submit three vests. The samples shall be representative of the construction, workmanship, components and materials to be used during production. When a contractor is in continuous production of these vests from contract to contract, submission of further first article inspection samples, on the new contract, may be waived at the discretion of the procuring activity (see 6.2f). Approval of the first article inspection samples or the waiving of the first article inspection does not preclude the requirements for performing the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2g).

4.4.2 First article test report. Upon completion of the first article inspection program (4.4), the government activity responsible for conducting the program (see 6.2g) shall report the results of the tests, with appropriate recommendations, to the contracting officer.

4.4.3 First article sample disposition. Upon successful completion of the first article inspection program, sample disposition will be as follows:

a. One sample will be returned to the manufacturer for use in monitoring production, and
b. Two samples will be destroyed during the first article inspection program and shall not be considered as part of the quantity to be delivered under the contract.

4.5 Quality conformance inspection. The sampling and inspection levels shall conform to MIL-STD-105 and Table VII.

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The quality conformance inspection shall be as specified in Table VII.

4.5.1 Sampling.

4.5.1.1 Inspection lot.

4.5.1.1.1 Vests. An inspection lot size shall be expressed in units of one vest made essentially under the same conditions and from the same materials and components. The sample unit shall be one vest.

4.5.1.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing vests, fully prepared for delivery from essentially the same materials and components. The sample unit shall be one shipping container, containing vests, fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for examinations and tests of the vests. The sample size, acceptance criteria, examinations, and tests required for the vests shall be as specified in Table VII.

4.6 Quality conformance verification.

4.6.1 Quality conformance verification inspection. The quality conformance verification inspection shall be as specified in Table VIII.

4.6.2 Quality conformance verification samples. Upon completion of the tests and examinations specified in Table VIII, a random sample shall be selected from each lot in accordance with MIL-STD-105, Inspection Level S-3. The tests and examinations required for the vests shall be as specified in Table VIII. The sample size and acceptance criteria shall be in accordance with MIL-STD-105 for single sampling plans for normal inspections. Each assembly, selected as a sample unit, shall be identified by its assigned serial number (see 3.5.1) and shall be forwarded to the inspection facility specified in the acquisition document (see 6.2). Additionally, the serial numbers of the units within the lot, represented by the sample units, shall be furnished to the inspection facility.

4.6.3 Quality conformance verification approval. Upon completion of the quality conformance verification inspection, the Government activity responsible for conducting the inspection program (see 6.2) shall report the results of tests, with recommendations, to the contracting officer of the procuring activity specified in the acquisition document. Final acceptance of the lot from which the samples were selected shall be based upon completion of the inspection program by the Government Quality Assurance Representative/Specialist.

4.7 Inspection methods.

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4.7.1 Visual examination. Each of the vests, selected as a sample from the lot, shall be visually examined to determine conformance to the requirements specified in 3.8.

4.7.2 Dimensional check. Each of the vests, selected as a sample from the lot, shall be dimensionally checked to determine conformance to the requirements specified in 3.9. The vest to be measured shall be laid flat, without tension, on a smooth, flat surface so that any creases or wrinkles shall not affect the measurement. The dimensions of the vest and components, location of the components, and width of the seams and hems shall be determined by a yardstick, metal tape, or by any suitable measuring device.

4.7.3 Packaging. Each of the fully prepared shipping containers, containing vests, selected as a sample unit from the lot, shall be visually examined to determine that the packaging, packing, and marking conform to the requirements specified in section 5 and 3.10.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2h).

5.1.1 Level A. Each vest, prior to being packaged, shall be fully zippered and the pockets shall be properly fastened. The vest shall be neatly folded in accordance with the commercial practice to an approximate size, 11-3/4 by 7-1/2 inches. A paper label shall be inserted (see 5.3.1) and each folded vest shall then be heat sealed within a clear polyethylene bag of 0.00125 +/- 20 percent inch thick material. The heat seal seams shall be straight, continuous, and parallel to each other and the edges of the polyethylene bag. The closing heat seal seam of the bag shall be as close as possible to the edge of the bag. One corner of each bag shall contain a hole, 1/4 inch in diameter, to allow the excess air to escape. As an alternate, the final closure of the bag may be accomplished by use of a tuck or reverse flaps. The approximate flat size of the polyethylene bag shall be 14 by 10 inches. No part of the vest shall be caught in a heat seal seam.

5.1.2 Commercial. Each vest assembly shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2h).

5.2.1 Level A. Twenty vests, preserved as specified in 5.1.1, shall be packed as specified in 5.2.2, except that the fiberboard container shall be Weather-Resistant Class, Variety SW, Grade V3c or V3s.

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5.2.2 Level B. Twenty vests, packaged as specified in 5.1.1, shall be packed flat, two in length, two in width, and five in depth, within a fiberboard shipping container conforming to MIL-B-17757, Type CF, Domestic Class, Variety SW, Grade 200, Size 3A. The top and bottom of the contents of each shipping container shall be completely covered with a sheet of 30 pound minimum basis weight kraft paper conforming to UUU-P-268, Grade B. The container shall be assembled, closed, and reinforced in accordance with MIL-B-17757, except that metal fasteners or stitches shall not be used. All the seams, corners, and manufacturer's joints shall be sealed with tape conforming to PPP-T-45, Type III, Grade A, B, or C. The body joint and the top and bottom flaps shall be glued together as specified in MIL-B-17757.

5.2.3 Commercial. The vests, preserved as specified in 5.1.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special markings required by the contract or purchase order (see 6.2i), the exterior shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.3.1 Polyethylene bag. Prior to heat sealing, a white paper label with legible and durable black letters and numerals shall be placed inside each polyethylene bag. The label shall be placed so that the information on the label shall be capable of being read through the bag. The paper shall contain the same information as specified in 5.3.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The vest covered by this specification is intended to be worn as part of the Aircrew Integrated Survival Armor Protection (AISAP) system for USN/USMC helicopter aircrewmen. It is intended to provide a modular pocket system and a platform for both soft and hard body armor.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, including any amendments.

b. Quantity and size desired.

c. Applicable drawings, including revisions.

d. Government part number and National Stock Number.

e. Issue of DODISS to be cited in the solicitation, and if

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required, the specific issue of individual documents referenced (see 2.1.1).

f. Whether first article inspection is waived (see 4.4.1).

g. Name and address of the first article inspection facility (see 4.4.1); and the name and address of the Government activity responsible for conducting the first article inspection program (see 4.4.2).

h. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

i. Whether any special markings are required (see 5.3).

j. Items of data required (see 6.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Paragraph	DID Number	DID Title	Tailoring
4.4.2	DI-NDTI-80809	Test reports	Use contractor format

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems Data Requirements Control List (AMSDL), must be reserached to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command, Department of the Navy, Washington, DC 20361; however, authorization for qualification of products may be obtained from the Commanding Officer, Naval

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Air Warfare Center Aircraft Division Warminster, (Code 6031), Warminster, PA 18974-5000.

6.5 First article. When first article inspection (see 3.2) is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.4.1. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Patterns and drawings. For access to patterns and drawings, address the acquiring activity issuing the invitation for bids.

6.7 Subject term (key word) listing.

Aircrewmen, helicopter
AISAP
Armor, body
Armor, integrated protection
Pocket, modular
Protection system

Preparing Activity
Navy-AS
(Project No. 8415-N963)

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Material	Nomenclature of component parts	Cut parts
Mesh (NAWCADWAR Purchase Description)	Mesh vest, left side	1
	Mesh vest, right side	1
Cloth, oxford, aramid (MIL-C-7219)	Adjustment strap cover flap	1
	Adjustment strap back-up piece	1
	Water Pocket, left side	1
	Water Pocket, right side	1
	Chem-lite pocket	2
	SRU-31/P medical pocket	2
	HEED pocket, cover	1
	HEED pocket, base	2
	Day/Night flare pocket	2
	Sea dye marker pocket	2
	Survival knife pocket	2
	Compass, whistle, and mirror pocket	1
	Pengun flare signal kit pocket	2
	SDU-5/E light storage pocket	2
	ANC/PRC radio pocket, panel-front	2
	ANC/PRC radio pocket, panel-back	2
	ANC/PRC radio pocket, side panel	1
	D-Ring retaining flap, right	1
	D-Ring retaining flap, left	1
	Reinforcement patch	2
	Water pocket liner	2
Cloth, duck, nylon (Validated) (MIL-C-3953)	Stiffner, chem light pocket	2
	Stiffner, HEED pocket	1
	Stiffner, day/night flare pocket	2
	Stiffner, sea dye marker pocket	2
	Stiffner, survival knife pocket	2
	Stiffner, pengun flare signal kit pocket, middle	1
	Stiffner, pengun flare signal kit pocket, top	1
	Stiffner, SRU-31/P medical pocket	2
	Stiffner, SDU-5/E light, storage pocket, front	1
	Stiffner, SDU-5/E light, storage pocket, back	1
	Stiffner, radio pocket	2

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TABLE II. Classification of defects for the visual examination of the vests

Defect	Classification	
	Critical	Minor
GENERAL		
a. Any non-specified hole, except drill holes, scissors, or knife cut, tear, mend, burn, or weakening defect such as smash, multiple float, loose slub, needle chew or abraded area	X	
b. Color of any component not as specified	1/	
c. Opposite sides of each main panel do not approximate each other in shade or any shade bar, spot, or stain		X
d. Any thread end not trimmed to a length of 1/4 to 3/8 inch or loose thread ends not removed		X
e. Any cut edge of the nylon cord, webbing, or tape, except for the hook and pile fastener tapes, not seared or any sharp edge formed by the searing	1/	
f. Any cut edge of the nylon duck not overedged stitched (serged)	1/	
g. Any cord or webbing used for the harness, belt, or ammunition keeper not fabricated from one continuous length.		X
h. Less than 25 percent of the hooks or pile flattened or missing from any fastener tape closure area		X
i. 25 percent or more of the hooks or pile flattened or missing from any fastener tape closure area	X	
j. Any hook or pile tape misplaced, not serving the intended purpose	X	
k. Any drill hole more than 1/16 inch in diameter or not covered by any attached component		X
METALLIC COMPONENTS		
a. Any surface rough, misaligned, or contains any crack, nick, burr, sharp edge, dent, or sliver	X	
b. Any surface unclean or contains embedded foreign matter		X

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TABLE II. Classification of defects for the visual examination of the vests - Continued

Defect	Classification	
	Critical	Minor
METALLIC COMPONENTS (Continued)		
c. Any loose, detached, or otherwise not securely retained.	X	
d. Any malformed, corroded, fractured, broken, chipped, bent, or distorted	X	
e. Any improperly finished or finish missing		X
f. Any rivet, snap fastener, or grommet misplace or improperly clinched resulting in cutting of the cloth, tape, or webbing, as applicable	X	
g. Any snap fastener mismated 2/		X
h. Any part of the slide fastener assembly bent, broken, slider not located at specified position, when the slide fastener is closed, or the movement of the slider is interfered with 3/	X	
i. Any unidirectional snap fastener with dot not located at specified position	X	
COMPONENTS AND ASSEMBLY		
a. Any component not specified or any defect of a component or defect of assembly not herein classified	1/	
b. Any component, component part, or required operation omitted or any operation improperly performed not herein classified	1/	
SEAMS AND STITCHING		
a. Accuracy of Seaming:		
1. Any seam, hem, or attachment of any component twisted, puckered, pleated, or caught in an unrelated operation or stitching; any seam edge not properly forced out or contains any fold		X
2. Ends of stitching that are not caught in other seams or stitching:		
(a) Not securely backstitched	X	
(b) Secure but backstitched for a distance less than 1/2 inch		X
(c) Backstitching not overlapped on itself		X

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TABLE II. Classification of defects for the visual examination of the vests - Continued

Defect	Classification	
	Critical	Minor
SEAMS AND STITCHING (Continued)		
3. Thread breaks:		
(a) Not securely overstitched	X	
(b) Secure but overstitched for a distance less than one inch		X
b. Gage of stitching 4/:		
1. Any stitching irregular, unevenly gaged, or various stitchings not uniform		X
c. Open seams 5/:		
1. Other than on any hem:		
(a) Up to an including 1/4 inch		X
(b) More than 1/4 inch but not more than one inch		X
(c) More than one inch	X	
2. On any hem:		
(a) 1/2 inch or less		X
(b) More than 1/2 inch		X
d. Raw edges 6/:		
1. More than 1/4 inch but not more than one inch		X
2. More than one inch		X
e. Seam and stitch type:		
1. Not specified seam or stitch	1/	
f. Stitch tension 7/:		
1. Loose tension resulting in a loose seam or tight tension resulting in breaking of the stitches, when normal pull is applied	X	
g. Stitches per inch 8/:		
1. Stitches more or less than specified: (a) One or 2		X
(b) Three		X

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TABLE II. Classification of defects for the visual examination of the vests - Continued

Defect	Classification	
	Critical	Minor
SEAMS AND STITCHING (Continued) (c) More than 3	X	
h. Bartacks:		
1. Any loose or misplaced, not serving the intended purpose		X
MARKINGS		
a. Any missing, incomplete, incorrect, or illegible		X
b. Improperly located		X

1/ The defect shall be considered critical when it seriously affects the serviceability or appearance, otherwise it is to be classified as a minor defect.

2/ The snap fasteners shall be checked for proper function and attachment by snapping closed and unsnapping each snap fastener at least three times.

3/ The slide fasteners shall be checked for proper function by opening and closing each slide fastener at least three times along its entire length.

4/ The gage of stitching defect shall be scored only when the condition exists on the major portion of any row of stitching.

5/ A seam shall be classified as open when one or more stitches, joining a seam, are broken or when two or more continuous skipped stitches or run-offs occur. On double-stitched seams, a seam shall be classified as open when either one or both sides of the seam contains any of the aforementioned defects for a single stitched seam.

6/ Any edge that should be turned under and is not, but is securely caught in the stitching, shall be classified as a raw edge. Any raw edge that is not securely caught in the stitching shall be classified as an open seam.

7/ Puckering is evidence of tight tension or gathering of the material. When puckering is evident, and not caused by the gathering of the material, the seam or hem, as applicable, shall be inspected by exerting normal pull in the direction of the stitching. Gathering of the material shall be classified as an accuracy of seaming defect.

8/ The stitches per inch defects shall be scored only when the condition exists on the major portion of any row of stitching.

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TABLE III. List of defects for the dimensions of the assembled vests

Examine	Defect
Measure the components, location of the components, width of the seams and hems, and length and width of the assembled vest	Any measurement deviating from the dimensions and tolerance as specified in the applicable drawings shall be enumerated as a dimensional defect.

TABLE IV. List of defects for packaging

Item	Defect
Exterior and interior marking	Missing, incomplete, incorrect, illegible; of improper size, location, sequence, or method of application; information on the label in the polyethylene bag not readable through the film or no label in the polyethylene bag
Materials	Any non-conforming component; any component or component part missing, damaged, or otherwise defective
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, container flaps, loose strapings, etc.; bulging or distortion of the containers; exterior containers contain metal fastenings or stitches; polyethylene bag vent hole missing, open or non-continuous heat sealed seams; any portion of the vest caught in a heat sealed seam.
Exterior and interior weight or content	Gross or net weight exceeds the requirements; number per container is more or less than specified; not individually packaged within the polyethylene bag; manner of folding damages the vest

TABLE V. Qualification inspection (see 4.3)

Inspection	Paragraph		Sample Size
	Requirement	Method	
Visual examination	3.8, applicable drawings & Table II	4.7.1	Every vest
Dimensional check	3.9, applicable drawings & Table III	4.7.2	Every vest
Preparation for delivery	3.10, section 5 & Table IV	4.7.3	Every vest

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TABLE VI. First article inspection (see 4.4)

Inspection	Paragraph		Sample Size
	Requirement	Method	
Visual examination	3.8, applicable drawings & Table II	4.7.1	Every vest
Dimensional check	3.9, applicable drawings & Table III	4.7.2	Every vest
Preparation for delivery	3.10, section 5 & Table IV	4.7.3	Every vest

TABLE VII. Quality conformance inspection (see 4.5)

Inspection 1/	Paragraph		Sample Size 2/	Acceptance Criteria 2/
	Requirement	Method		
Visual examination	3.8, applicable drawings & Table II	4.7.1	Every vest for critical defects. Inspection Level II for minor defects.	Reject all units with any critical defects. An acceptable quality level off 15.0 defects per 100 units for minor defects.
Dimensional check	3.9, applicable drawings & Table III	4.7.2	Inspection Level S-3	An acceptable quality level of 4.0 defects per 100 units
Preparation for delivery	3.10, Section 5 & Table IV	4.7.3	Inspection Level S-2	An acceptable quality level of 2.5 defects per 100 units

1/ The results of the examinations and tests shall be identifiable by the assigned serial numbers (see 3.5.1).

2/ The sample size and acceptance criteria shall be in accordance with MIL-STD-105.

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Table VIII. Quality conformance verification inspection (see 4.6)

Inspection 1/	Paragraph		Sample Size 2/
	Requirement	Method	
Visual examination	3.8, applicable drawings & Table II	4.7.1	Every vest
Dimensional check	3.9, applicable drawings & Table III	4.7.2	Every vest
Preparation for delivery	3.10, Section 5 & Table IV	4.7.3	Every vest

1/ The results of the examinations and tests shall be identifiable by the assigned serial numbers (see 3.5.1)

2/ The sample size and acceptance criteria shall be in accordance with MIL-STD-105

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1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-V-29601(AS)	2. DOCUMENT DATE (YYMMDD) 941215

3. DOCUMENT TITLE
VEST, SURVIVAL, CMU-30/P22P-15

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial: (2) DSN: (*If Applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION	b. TELEPHONE NUMBER (*Include Area Code*) (1) Commercial: (908) 323-7488 (2) DSN: 624-7488
c. ADDRESS (*Include Zip Code*) CODE 4.1.11.2B120-3 HIGHWAY 547 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office, 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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